

Chapter 290, Health and Safety Services

Section 56, Chemical Safety

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Responsible Department: Environmental Health & Safety (EH&S)

Source Document: Policy on Management of Health, Safety, and the Environment

I. Purpose

- A. This section outlines policy and responsibilities for the safe use, storage, transportation, shipping, and disposal of chemicals in research, teaching, or other activities to minimize risks of illness or injury. It also provides reporting procedures for emergencies and chemical theft.
- B. For requirements on radioactive materials, see [Section 290-75](#) and for controlled substances, see [Section 290-70](#).
- C. Additional requirements for UC Davis Health are outlined in UC Davis Health Administrative Policies and Procedures Sections [1605](#), [1612](#), [1620](#), [1623](#), [1641](#), [1646](#), and [1648](#).

II. Policy

- A. The University manages chemicals in a manner that safeguards:
 - 1. The health and safety of all faculty/academics, staff, students, employees, volunteers, and visitors.
 - 2. Wild and domestic animals on UC Davis properties or in surrounding areas.
 - 3. The environment.
- B. Reporting Emergencies and Theft
 - 1. Emergencies involving chemicals must be reported immediately to:
 - a. At Davis: Dial 9-1-1 for the UC Davis Fire Department.
 - b. At UC Davis Health: Dial 9-1-1 for Sacramento Fire Department (see UC Davis Health Administrative Policy Section [1605](#)).
 - c. At off-site locations: Dial 9-1-1 for local emergency services.
 - 2. Theft or diversion of [Chemicals of Interest](#) (COI) must be reported immediately to the [UC Davis Police Department](#) and Environmental Health & Safety (EH&S).

III. Requirements

- A. Use and Storage
 - 1. Consumption of food, beverage, medicines, tobacco, chewing gum, and cosmetics application, as well as handling contact lenses, is prohibited in areas where hazardous chemicals are used or stored.
 - a. New laboratory buildings must include designated eating areas for employees.
 - b. Departments must provide employee designated areas to eat and store food outside of research and teaching laboratories where hazardous chemicals, biological materials, or radiological materials are used or stored.
 - 2. Hazardous materials (e.g., flammable chemicals, highly reactive or explosive chemicals, acutely toxic gases, cryogenics) must comply with [Chemical Hygiene Plan](#) (CHP).
 - 3. Chemical explosives must comply with applicable Federal, State, local laws, and regulations, as well as University policies, procedures, and standards.

4. Chemical storage must comply with the California Fire Code, California Building Code, and National Fire Protection Association (NFPA) standards concerning hazardous chemicals (see [Section 390-40](#) and the CHP).
 5. Chemicals transferred from their original containers or produced in laboratory procedures must be properly identified and labeled per the CHP.
 6. Use of Chemical Fume Hoods and Biological Safety Cabinets
 - a. Volatile, toxic, or gaseous chemicals must be handled in a certified chemical fume hood and used per [SafetyNet #35](#) and the CHP.
 - b. If such chemicals must be used in a bio-safety cabinet, it must be a Type II, Class B1, B2, or B3 with exhaust vented outside the building.
- B. Transportation and Shipping
1. Transporting dangerous goods (including hazardous chemicals) on public roads must comply with U.S. Department of Transportation and the California Departments of Agriculture, Health, and Highway Patrol requirements.
 2. Shipments of dangerous goods must comply with all U.S. Department of Transportation and International Air Transport Association (IATA) requirements.
 3. It is strongly recommended that employees use University vehicles when transporting hazardous chemicals in quantities under the Department of Transportation Materials of Trade exception limit. See [49 CFR Chapter 1, 173.6](#).
 4. Passenger compartment transport is prohibited, except for small quantities in original, unopened UN-specified packing as shipped from the vendor.
 5. Packages containing hazardous chemicals must be secured during transit to prevent movement or damage.
- C. Hazardous Waste Management
1. Individuals creating and managing hazardous waste must successfully complete training on the correct procedures for managing hazardous waste.
 2. Waste disposal must be handled through EH&S. Off-site hazardous waste disposal contractors must be approved by EH&S for all on-campus facilities.
 - a. EH&S manages hazardous waste manifests and maintains shipment records for on-campus facilities.
 - b. Off-campus facilities must coordinate with EH&S for waste disposal contracting support.
 3. Hazardous waste containers must be labeled per [SafetyNet #110](#).
 4. All containers must be securely sealed and stored in appropriate secondary containment.

IV. Roles and Responsibilities

A. [UC Davis Chemical & Laboratory Safety Committee \(CLSC\)](#)

Develops, facilitates, and implements policies and procedures for the safe acquisition, storage, use, and disposal of chemicals.

B. [Environmental Health & Safety \(EH&S\)](#)

1. Assists with the developing and implementing the CHP and [Hazard Communication Program](#).
2. Provides employees with hazardous chemicals safety information and training (see [Section 290-27](#)).

3. Evaluates hazardous chemical use and recommends control methods.
4. Collaborates with Fire Prevention Services to provide guidance on the safe use and storage of hazardous chemicals.
5. Manages the online [Chemical Inventory System](#), including preparation and submission of Community Right-To-Know hazardous materials inventory.
6. Conducts laboratory safety reviews, and chemical exposure monitoring.
7. Works with UC Davis Police Department to assist personnel in [Chemical Facility Anti-Terrorism Standard \(CFATS\)](#) compliance.
8. Recommends protective clothing and equipment (see [Section 290-50](#)).
9. Oversees the integrated [Hazardous Waste Program](#).
10. Collaborates with Supply Chain Management to ensure the safe transportation of hazardous chemicals.

C. Department Heads/Chairs

1. Establish, maintain, and support safety programs to ensure a safe and healthy work environment for employees, students, visitors, and volunteers per Sections [290-15](#), [290-27](#), and CHP or Hazard Communication Program.
2. Departments or units using hazardous chemicals in a laboratory must:
 - a. Implement the CHP and Hazardous Chemicals Communication Program (see [Section 290-27](#)) as part of their Injury and Illness Prevention Program (IIPP) (see [Section 290-15](#)).
 - b. Maintain current chemical inventory in the Chemical Inventory System and update the inventory once a year at a minimum.
 - c. Record and maintain all COI as defined in CFATS in the Chemical Inventory System, regardless of quantity (see [Section 390-60](#)).

D. Department Safety Coordinators

1. Fulfill the responsibilities outlined in [Section 290-15](#) or UC Davis Health Administrative Policy Section [1620](#).
2. Assist in implementing the CHP or Hazard Communication Program.
3. Provide employees with hazardous chemicals safety training (see [Section 290-27](#)).
4. Annually review and update the [Injury and Illness Prevention Program](#) (IIPP), and facilitate annual training for all personnel.
5. Assist Principal Investigators (PIs) in obtaining required authorizations.

E. Principal Investigators (PIs) and Supervisors

1. Implement and maintain safety programs for employees, students, visitors, and volunteers.
2. Fulfill the responsibilities outlined in [Sections 290-15](#), [290-27](#), and [290-50](#).
3. Fulfill the responsibilities outlined in the CHP or Hazard Communication Program, and implement the plan as applicable.
4. Maintain a current chemical inventory in the Chemical Inventory System.
5. Ensure employees and students comply with mandated medical monitoring programs and are aware of voluntary medical monitor programs, consistent with [Section 290-60](#).
6. Maintain training records for at least 3 years and carcinogen exposure records for at least

30 years per [29 CFR §1910.1020](#).

7. Enforce prohibited activities (e.g., eating, drinking, tobacco use, cosmetic application, or handling contact lenses) in areas where hazardous chemical, biological, or radiological materials are used or stored.
8. Researchers in leased spaces should refer to their rental agreement for maintenance and annual services responsibilities.

F. Employees, students, visitors, and volunteers

1. Individuals handling, using, storing, transporting, or disposing of hazardous chemicals must:
 - a. Complete chemical spill response and disposal training(see [SafetyNet #13](#)).
 - b. Wear appropriate personal protective equipment (see [Section 290-50](#)).
2. Fulfill the responsibilities outlined in [Sections 290-15](#), [290-27](#), and [290-50](#).
3. Fulfill the responsibilities outlined in the CHP or Hazard Communication Program.
4. Successfully complete required safety training. See the [Safety Training Matrix for Laboratory Personnel](#).
5. Comply with the [hazardous waste disposal](#) and [spill procedures in SafetyNet#13](#).
6. Participate in mandatory medical surveillance, if applicable.

G. [Occupational Health Services](#)

1. Provides medical surveillance for employees working with chemical carcinogens and other hazardous chemicals.
2. Collaborates with the Student Health and Wellness Center to provide medical surveillance for students who work with chemical carcinogens or other hazardous chemicals.

H. [Student Health and Wellness Center](#)

Collaborates with Occupational Health Services to provide medical surveillance for students working with chemical carcinogens or other hazardous chemicals.

I. [Facilities Management](#) / Plant Operation and Maintenance (PO&M)

1. Certifies and maintains chemical laboratory and storage area fume hoods, ventilation systems, pollution control equipment, eyewash/emergency showers, and other engineering control safety devices.
2. Conducts annual safety inspections or more frequently checks if needed to ensure their proper operation and compliance to regulatory requirements.

J. [Supply Chain Management](#)

Coordinates the shipping for all hazardous, chemical, and radiological material.

K. UC Davis [Fire Department](#) and [Fire Prevention Services](#)

Fulfill responsibilities outlined in [Section 390-40](#).

L. UC Davis [Police Department](#)

Fulfill responsibilities outlined in [Section 390-60](#).

V. Further Information

- A. For further information and assistance, contact [EH&S](#) (Davis, 530-752-1493; UC Davis Health, 916-734-2740).

1. UC Davis [Chemical Hygiene Plan \(CHP\)](#).
 2. UC Davis [Hazard Communication Program](#).
 3. UC Davis [Hazardous Waste Program](#).
 4. UC Davis [Injury & Illness Prevention Program \(IIPP\)](#).
- B. Additional guidance on storing and using hazardous explosives is available from the [UC Davis Fire Department](#) and [EH&S](#).
- C. Additional assistance on transporting hazardous chemicals is available from chem-safety@ucdavis.edu.

VI. References and Related Policies

- A. [U.S. Code of Federal Regulations](#):
1. [29 CFR §1910.1020](#), Access to employee exposure and medical records.
 2. [49 CFR](#), Department of Transportation:
 - a. [Chapter I](#), Pipeline and Hazardous Materials Safety Administration.
 - b. [Chapter III](#), Federal Motor Carrier Safety Administration, Department of Transportation.
- B. [National Fire Protection Association](#) (NFPA).
- C. [Chemical Facility Anti-Terrorism Standard](#) (CFATS).
- D. [International Air Transport Association](#) (IATA), [Dangerous Goods Regulations](#).
- E. [California Health and Safety Code](#), Division 20, Miscellaneous Health and Safety Provisions.
1. [Chapter 6.5](#), Hazardous Waste Control.
 2. [Chapter 6.95](#), Hazardous Materials Release Response Plans and Inventory.
- F. [California Code of Regulations](#):
1. [8 CCR §3203](#), Injury and Illness Prevention Program.
 2. [8 CCR §3204](#), Access to Employee Exposure and Medical Records.
 3. [8 CCR §3220](#), Emergency Action Plan.
 4. [8 CCR §3221](#), Fire Prevention Plan.
 5. [8 CCR §3380](#), Personal Protective Devices.
 6. [8 CCR §5191](#), Occupational Exposure to Hazardous Chemicals in Laboratories.
 7. [8 CCR §5192](#), Hazardous Waste Operations and Emergency Response.
 8. [8 CCR §5194](#), Hazard Communication.
 9. [22 CCR Division 4.5 §66260-69600](#), Environmental Health Standards for the Management of Hazardous Waste.
- G. [California Fire Code](#).
- H. [California Building Standards Code](#).
- I. Office of the President:
1. [Management of Health, Safety, and the Environment](#).
 2. [Personal Protective Equipment](#).
- J. UC Davis EH&S [SafetyNets](#):

1. [SafetyNet #13, Guidelines for Chemical Spill Control.](#)
 2. [SafetyNet #35, How to Use a Chemical Fume Hood Safely.](#)
 3. [SafetyNet #110, Guidelines for Completing the Chemical Waste Label.](#)
- K. UC Davis Policy and Procedure Manual:
1. Section [290-15](#), Safety Management Program.
 2. Section [290-27](#), Hazardous Chemicals Communication Program.
 3. Section [290-50](#), Protective Clothing and Equipment.
 4. Section [290-60](#), Occupational and Preventive Medicine.
 5. Section [290-70](#), Controlled Substances.
 6. Section [290-75](#), Radiological Safety--Health Physics.
 7. Section [290-95](#), Pesticide Applications.
 8. Section [350-20](#), Procurement and Use of Tax-Free Alcohol.
 9. Section [390-10](#), Campus Emergency Policy.
 10. Section [390-40](#), Fire Safety.
 11. Section [390-60](#), Laboratory Security.